Technical Report: Library Management System

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1. Introduction

The Library Management System is a robust web application designed to streamline the management of book records within a library environment. This project was developed as part of the Software Engineering Internship assignment for Expernetic LLC. The primary goal was to implement a full-stack solution enabling users to perform core CRUD (Create, Read, Update, Delete) operations on book data while ensuring high performance and a user-friendly interface.

2. Project Scope

The scope of this project includes:

- **Backend Development**: Creating a RESTful API using C# and .NET 10.
- **Database Integration**: Implementing an SQLite database with Entity Framework Core for data persistence.
- **Frontend Development**: Building a responsive and interactive user interface using React and TypeScript.
- **Authentication**: Implementing a secure user registration and login system (Optional feature included).
- **Search Functionality**: Allowing users to filter books by title or author.

3. Technical Architecture

3.1 Backend Implementation

The backend is built following the **Clean Architecture** principles to ensure maintainability and scalability.

- **Framework**: ASP.NET Core Web API.
- **Data Access**: Entity Framework Core was utilized to interact with the **SQLite** database.
- **Validation**: Server-side validation was implemented using **Data Annotations** in Models and DTOs to ensure data integrity.
- **Error Handling**: Global exception handling with `try-catch` blocks and proper HTTP status codes ensures the API remains stable under various conditions.

3.2 Frontend Implementation

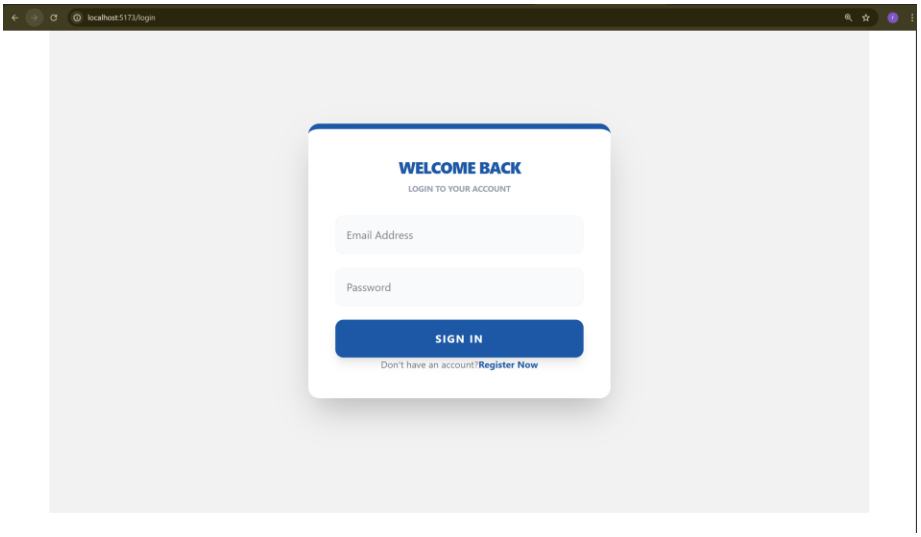
The frontend focuses on providing a seamless **User Experience (UX)**.

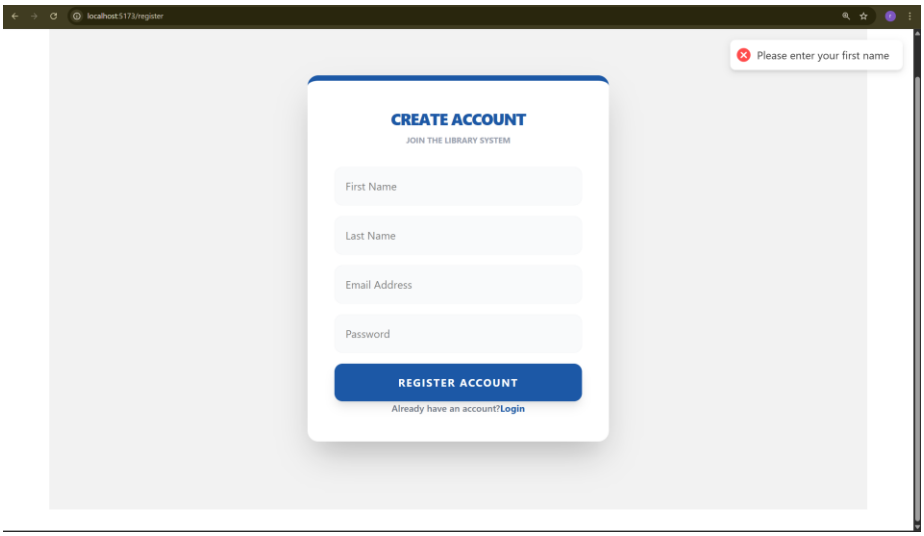
- **Framework**: React with **TypeScript** for strong typing and reduced runtime errors.
- **Styling**: **Tailwind CSS** was used for a modern, responsive design.
- **State Management**: React Hooks (`useState`, `useEffect`) were employed for efficient data handling.
- **Routing**: **React Router** manages navigation, including **Protected Routes** to restrict unauthorized access to administrative features.

- **Notifications**: **React Hot Toast** was integrated for real-time visual feedback on user actions (Like "Book Added Successfully").

4. Screenshots of the System

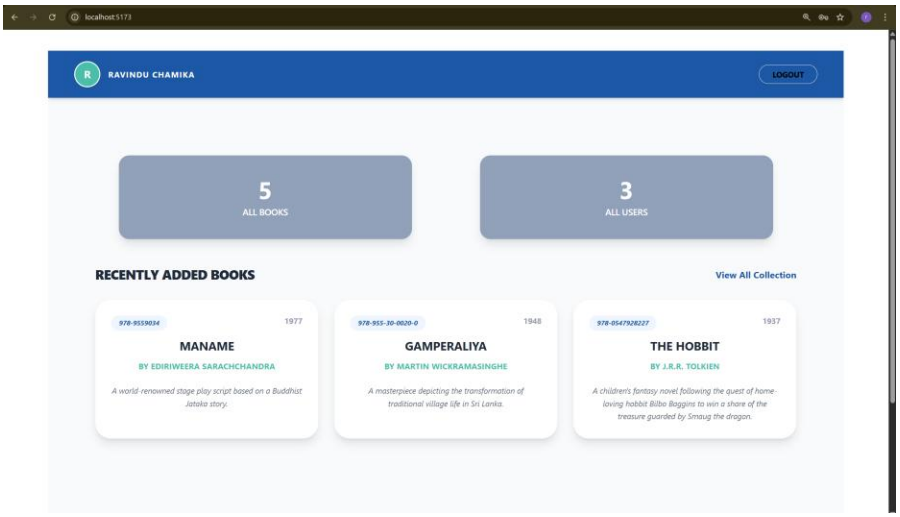
4.1 User Authentication (Login & Registration)

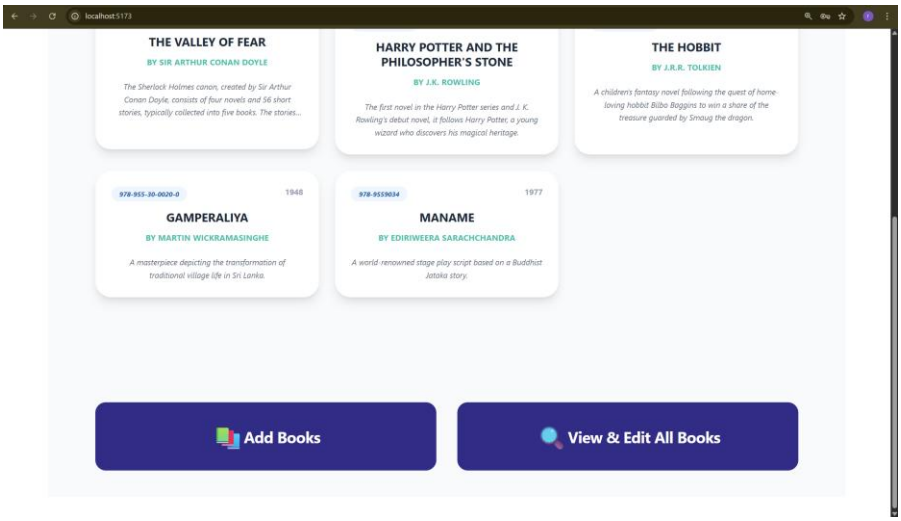




Description: Secure login interface for authorized library personnel.

4.2 Library Inventory (Book List & Search)


localhost:5173
RAVINDU CHAMIKA
LOGOUT
5
ALL BOOKS
3
ALL USERS
RECENTLY ADDED BOOKS
View All Collection
978-9559034
1977
MANAME
BY EDIRIWEERA SARACHCHANDRA
A world-renowned stage play script based on a Buddhist Jataka story.
978-955-30-0020-0
1948
GAMPERALIYA
BY MARTIN WICKRAMASINGHE
A masterpiece depicting the transformation of traditional village life in Sri Lanka.
978-0547928227
1937
THE HOBBIT
BY J.R.R. TOLKIEN
A children's fantasy novel following the quest of home-loving hobbit Bilbo Baggins to win a share of the treasure guarded by Smaug the dragon.


localhost:5173
THE VALLEY OF FEAR
BY SIR ARTHUR CONAN DOYLE
The Sherlock Holmes canon, created by Sir Arthur Conan Doyle, consists of four novels and 56 short stories, typically collected into five books. The stories...
HARRY POTTER AND THE PHILOSOPHER'S STONE
BY J.K. ROWLING
The first novel in the Harry Potter series and J. K. Rowling's debut novel, it follows Harry Potter, a young wizard who discovers his magical heritage.
THE HOBBIT
BY J.R.R. TOLKIEN
A children's fantasy novel following the quest of home-loving hobbit Bilbo Baggins to win a share of the treasure guarded by Smaug the dragon.
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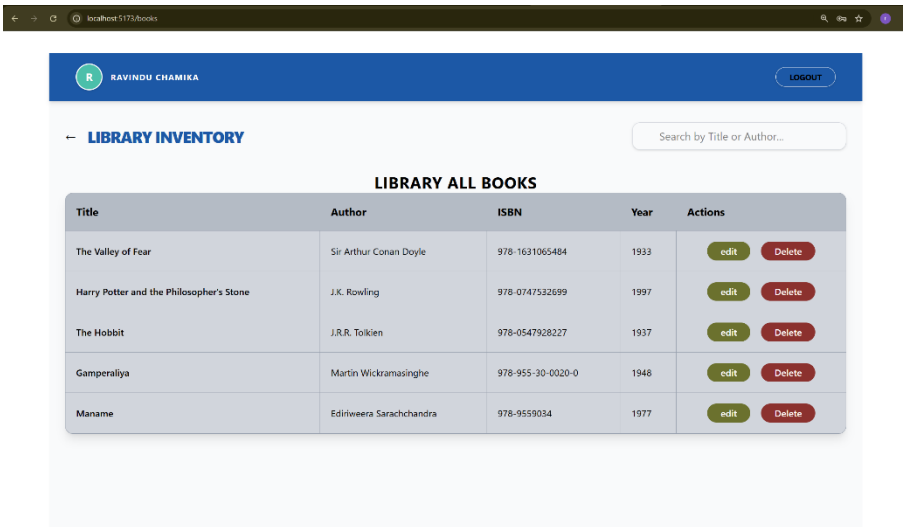
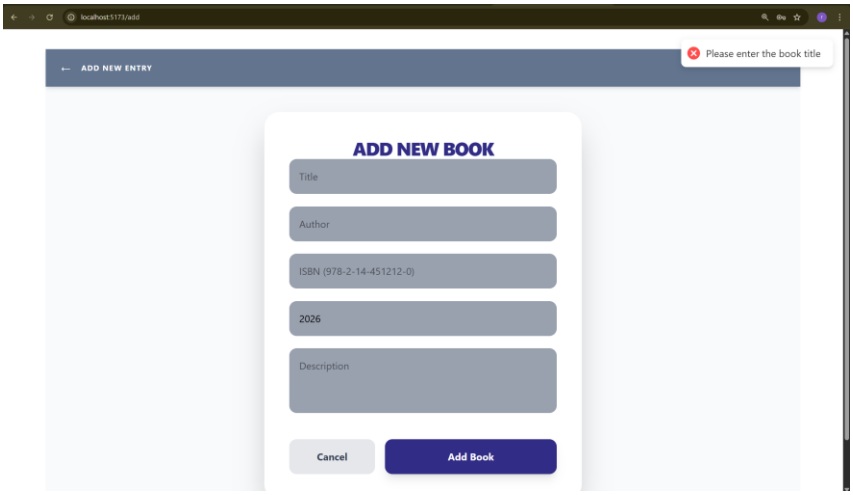
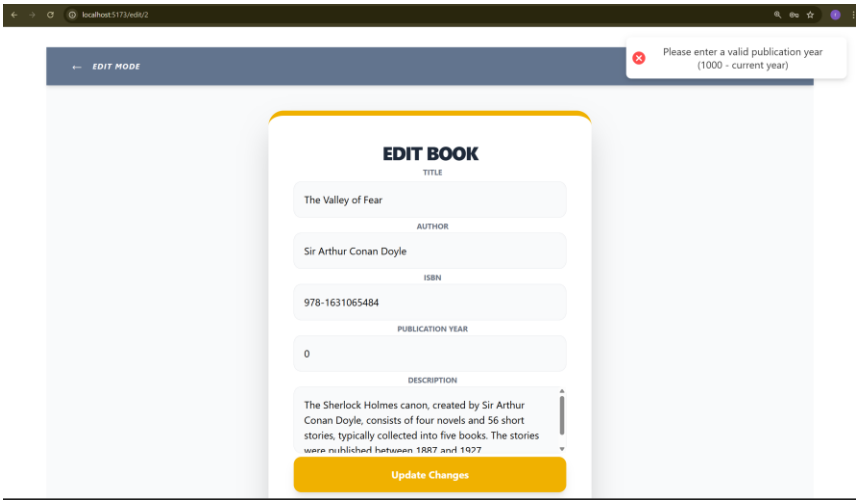
A world-renowned stage play script based on a Buddhist Jataka story.
Add Books
View & Edit All Books


localhost:5173/books
RAVINDU CHAMIKA
LOGOUT
LIBRARY INVENTORY
Search by Title or Author...
LIBRARY ALL BOOKS
Title
Author
ISBN
Year
Actions
The Valley of Fear
Sir Arthur Conan Doyle
978-1631065484
1933
edit
Delete
Harry Potter and the Philosopher's Stone
J.K. Rowling
978-0747532699
1997
edit
Delete
The Hobbit
J.R.R. Tolkien
978-0547928227
1937
edit
Delete
Gamperaliya
Martin Wickramasinghe
978-955-30-0020-0
1948
edit
Delete
Maname
Ediriweera Sarachchandra
978-9559034
1977
edit
Delete

Description: Displaying all books with real-time search and filtering capabilities.

4.3 Adding/Editing Records (With Toast Notifications)





Description: User friendly form featuring multi-layer validation and real-time feedback.

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5. Challenges Faced & Learning Curve

5.1 Adapting to a New Tech Stack

One of the most significant challenges during this project was transitioning to a new technology stack within a limited one-week timeframe.

- **Transition from Java/Spring Boot**: Although my core expertise lies in **Java and Spring Boot**, this assignment required a solution using **C# and .NET 10**. Leveraging

transferable skills in OOP and MVC architecture, I rapidly adapted to .NET-specific syntax, middleware configurations, and EF Core.
- **Adopting TypeScript**: Moving from standard JavaScript to **TypeScript** required a disciplined approach to defining interfaces and types. This was a critical step in reducing runtime errors and improving code maintainability.

5.2 Validation Synchronization

Ensuring that the backend's strict validation rules (like ISBN regex patterns) were correctly mapped to frontend toast notifications was crucial for a professional user experience. I implemented manual validation logic on the frontend to provide immediate feedback, backed by a secondary layer of validation on the server side.

6. Conclusion

Developing the Library Management System provided valuable insights into the seamless integration of .NET and React ecosystems. By focusing on robust error handling, multi-layer data validation, and responsive design, the application meets all requirements specified by Expernetic LLC while offering an aesthetically pleasing experience for the end-user.

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7. Appendix: How to Run the Application

To run this project locally, follow these steps:

7.1 Prerequisites

- .NET 10 SDK
- Node.js (v18 or higher)
- Git

7.2 Backend Setup

1. Navigate to the `LibraryManagementSystemBackend` folder.
2. Run migrations to setup the SQLite database:
    - `dotnet ef database update`
3. Start the API server:
    - `dotnet run`

    The backend will be available at: http://localhost:5089

7.3 Frontend Setup

1. Navigate to the LibraryManagementSystemFrontend folder.
2. Install dependencies:

    - npm install

3. Start the development server:

    - npm run dev

    The frontend will be available at: http://localhost:5173